LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

SAUERS

We show that not every countable group is MF, where MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. More precisely, the simple property-(T) group $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ has trivial image under every homomorphism to an MF group; consequently its reduced group C^*-algebra is a separable stably finite C^*-algebra that is not MF. The proof rests on a one-sided compression criterion for Kazhdan subgroups in norm matrix coronas. We also prove an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona, and use it to show that recognizing MF groups from finite presentations is Π^0_2-complete.

DRAFT — NOT READY FOR PUBLIC CONSUMPTION

This manuscript is actively being revised and has not been released for public consumption.

1. Introduction

Not every countable group is MF. Our counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)).$$

The group H is countable, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is not MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group G is *MF* if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

2020 *Mathematics Subject Classification.* Primary 46L05; Secondary 20C07, 16S50, 20G35.

Key words and phrases. MF groups, norm matrix coronas, MF radicals, property (T), asymptotic representations, Leavitt algebra.

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum.

A separable C^*-algebra is called MF if it embeds as a C^*-subalgebra of a norm matrix corona. For every countable group G, the reduced group algebra $C^*_{\mathrm{r}}(G)$ is separable and has a faithful canonical trace, hence is stably finite. Suppose that $e\colon C^*_{\mathrm{r}}(G) \to \mathcal{Q}_{\mathbf{d}}$ is an MF embedding and put $p = e(1)$. Then $g \mapsto e(\lambda_g) + (1-p)$ is an injective homomorphism from G into $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$, so G is MF. Therefore a non-MF group automatically gives a separable stably finite reduced group C^*-algebra that is not MF.

For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. Proposition 2.1 shows that G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

For a subgroup $L \le G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \le L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \le G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \le \mathfrak{D}_G(L)$, then*

$$K \le \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) *and $\mathfrak{D}_G(L) = G$, then* $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

We sketch the argument. Corollary 2.7 gives $\mathfrak{D}_G(L) \le R_{\infty\to 2}(G)$. That is, every operator norm asymptotic representation of G is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm. If a corona homomorphism were nontrivial on K, the Kazhdan projection of K would cut out a G-invariant corner on which K has no nonzero fixed vectors (Lemma 2.8); by the Kazhdan inequality some $s_0 \in K$ would then stay bounded away from the identity in normalized Hilbert–Schmidt norm along a subsequence, contradicting $s_0 \in \mathfrak{D}_G(L)$.

Theorem 2.3 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

We next apply Theorem A to the group H.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is countable, nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \mathrm{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1t_1)s_1 = 1$ shows that $d \neq 1$. Since H is simple, d normally generates H. Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)^{-1}$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \mathrm{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.6.

We also determine the arithmetical complexity of MF recognition for finite presentations, in the Kleene–Mostowski hierarchy whose levels Σ^0_n and Π^0_n count alternations of quantifiers over a decidable relation.

Theorem C (recognizing MF groups)**.** *Fix an effective coding of finite presentations of groups by natural numbers. The set of codes of finite presentations of MF groups is Π^0_2-complete, and the set of codes of finite presentations of non-MF groups is Σ^0_2-complete. In particular, neither set is recursively enumerable, and no algorithm decides from a finite presentation whether the presented group is MF. More precisely, there is a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the group presented by $\widehat{R}_e$ is MF if and only if the e-th partial computable function has infinite domain.*

Proposition 4.3 gives a Π^0_2 description of the MF presentations: a presentation is MF exactly when finite certificates exist at every scale (Section 4). For hardness we reduce the Π^0_2-complete set of programs with infinite domain to MF recognition: from e we construct a finite presentation $\widehat{R}_e$ (Section 6). If the domain is finite, then $H \leq \widehat{R}_e$, so $\widehat{R}_e$ is not MF. If the domain is infinite, then $\widehat{R}_e$ satisfies the hypotheses of Theorem 5.2, an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona, and is MF (Sections 5 and 7). The MF property is closed in the space of marked groups (Lemma 4.2; equivalently, it is closed under direct limits, by Korchagin [Kor, Proposition 13]), so every finitely generated group that is not MF is a quotient of a finitely presented one; Theorem C gives the uniform family and the exact level. Proposition 2.2 records the pullback formula $\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q))$ for surjections $f\colon G \to Q$ with $\ker f \leq \mathrm{Rad}_{\mathrm{MF}}(G)$.

Relation to prior work. Blackadar and Kirchberg introduced the notions of MF and NF algebras in a 1994 Oberwolfach abstract [BK94]. They wrote that the classes of NF, strong NF, and separable nuclear stably finite C^*-algebras might coincide. Their 1997 paper developed these notions and proved that a separable C^*-algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite C^*-algebra is quasidiagonal. This was a question about nuclear C^*-algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite C^*-algebra were MF, then every countable group would be MF because its reduced group C^*-algebra is separable and stably finite. Later papers on approximation of groups relate the resulting operator norm problem to Kirchberg's question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite C^*-algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group C^*-algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that $A *_C A$ is MF whenever A is a separable MF C^*-algebra and $C \subseteq A$ is a C^*-subalgebra [Sh, Theorem 10]. She also proves that the full group C^*-algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten p-norms with $1 < p < \infty$ [DGLT, LO]. Bachner–Dogon–Lubotzky later proved nonapproximability for $p = 1$ [BDL]. The operator norm case $p = \infty$ remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]; that is the route taken here.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. The compression used here comes from their defining relations. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of GL_n over an exchange ring that are normalized by EL_n [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality $Z(R) = \mathbb{F}_2$, implies that $\mathrm{EL}_{12}(R)$ is simple (Proposition 3.3). Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property (one that some finitely presented group has and that no group containing some fixed finitely presented group has) can be recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov, and Adian–Rabin gives undecidability; Theorem C gives the exact level. A classical Π^0_2-complete recognition problem for finitely presented groups is torsion-freeness, by Lempp [Lem]. For recursively presented groups, Bilanovic–Chubb–Roven proved that every Markov property is Π^0_2-hard to recognize [BCR]; their construction is the first step of Section 6. The passage from recursive to finite presentations is where the operator algebra enters: a generic Higman embedding does not preserve the MF property, and the HNN extensions of Section 6 are chosen so that Theorem 5.2 applies to them.

2. One-sided compression and the MF radical

This section proves Theorem A. Let $uLu^{-1} \leq L$. In a finite-dimensional representation ρ, conjugation by $\rho(u)$ maps the commutant of $\rho(L)$ into itself; the map is injective, hence surjective, so the commutant is invariant (Theorem 2.3). In a norm matrix corona the same conclusion is obtained for the Kazhdan projection of a property-(T) subgroup: conjugation moves the projection below itself, and stable finiteness of the corona forces equality (Section 2.3). A normal property-(T) subgroup contained in the compression defect then lies in the MF radical (Theorem 2.9).

2.1. **The MF radical.**

For $N \trianglelefteq G$ define its *MF kernel closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap \{\ker f : N \leq \ker f,\ f\colon G \to M,\ M \text{ MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$. This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

Proposition 2.1 (basic properties of the MF radical)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G, $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, and π is a corona homomorphism from G, then $\pi \circ \alpha$ kills x. Hence $\pi(\alpha(x)) = 1$ for every π, which proves full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$. If G/R is trivial, there is nothing to prove. If $x \in G/R$ is nontrivial and $g \in G$ maps to x, then $g \notin R$. Some homomorphism from G to an MF group therefore separates g from the identity. It kills R and descends to a homomorphism from G/R that separates x. Thus G/R is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

For $N \trianglelefteq G$, homomorphisms from G/N to MF groups correspond to homomorphisms from G to MF groups that kill N. It follows directly that

$$\mathrm{Rad}_{\mathrm{MF}}(G/N) = \mathrm{cl}^G_{\mathrm{MF}}(N)/N.$$

Since a countable group is MF exactly when its MF radical is trivial, this identity gives the asserted closure criterion. □

Proposition 2.2 (full-kernel pullback)**.** *Let $f\colon G \to Q$ be a surjective homomorphism of countable groups. If $\ker f \le \mathrm{Rad}_{\mathrm{MF}}(G)$, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

In particular, the conclusion holds whenever $\mathrm{Rad}_{\mathrm{MF}}(\ker f) = \ker f$. More generally, for every normal subgroup $N \trianglelefteq G$,

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = f^{-1}\Big(\mathrm{cl}^Q_{\mathrm{MF}}(f(N))\Big).$$

Consequently,

$$G/N \text{ is MF} \iff \ker f \le N \text{ and } Q/f(N) \text{ is MF}.$$

Proof. Every corona homomorphism from G kills $\ker f$ and therefore factors uniquely through f. Conversely, composing a corona homomorphism from Q with f gives one of G. Intersecting these two corresponding families of kernels gives the displayed MF radical pullback. For the stated sufficient condition, let $\pi\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be any corona homomorphism. Its restriction to $\ker f$ is trivial because $\mathrm{Rad}_{\mathrm{MF}}(\ker f) = \ker f$. Hence $\ker f \le \mathrm{Rad}_{\mathrm{MF}}(G)$.

The same factorization, restricted to homomorphisms that kill N, gives the closure formula. If this closure equals N, then it contains $\ker f$. Mapping the equality onto Q shows that $\mathrm{cl}^Q_{\mathrm{MF}}(f(N)) = f(N)$. Conversely, these two conditions give

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = f^{-1}(f(N)) = N.$$

The final equivalence now follows from Proposition 2.1, applied in G and Q. □

2.2. **Finite dimension.**

Theorem 2.3 (finite-dimensional commutant rigidity)**.** *Let G be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore $\rho([ucu^{-1}, \ell]) = 1$ for every $\ell \in L$. □

The finite-dimensionality hypothesis cannot be dropped: an injective linear self-map of an infinite-dimensional commutant need not be surjective.

2.3. **Kazhdan transport in normalized Hilbert–Schmidt norm.**

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator norm asymptotic representations of G.

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 2.4. *For every sequence of positive integers (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C})/\bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^* x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^* x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^*-algebra, equivalent projections $p \le q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1 - q)$ is an isometry in A. Consequently $ww^* = q$, and hence $p = q$. □

Lemma 2.5 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\mathrm{Fix}\,\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\,\mathrm{Fix}\,\pi(L) \subseteq \mathrm{Fix}\,\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. □

Theorem 2.6 (one-sided Kazhdan transport)**.** *Let $L \le G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \le L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\mathrm{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$

defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\mathrm{Ad}(A) - \mathrm{Ad}(B)\| \le 2\,\|A - B\|\,,$$

so the operator norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\mathrm{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2. The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$. The projection identity below is proved inside $\mathcal{B}$; the representation of $\mathcal{B}$ on $\mathcal{K}_\omega$ is used only afterwards. Since $uLu^{-1} \le L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.5, and hence $U^*PU \le P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.4 therefore gives $U^*PU = P$, so U commutes with P. Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\mathrm{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. □

Corollary 2.7. *Let $L \le G$ have property* (T)*, let $uLu^{-1} \le L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.6, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\Big\| V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1}) \Big\| \longrightarrow 0.$$

Therefore

$$\Big\| V_n([ucu^{-1}, \ell]) - 1 \Big\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. □

2.4. **Normal Kazhdan subgroups and the MF radical.** Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona homomorphism of G. Normality of K implies that p commutes with the image of G. Hence $q = 1 - p$ also commutes with that image, and in every nondegenerate representation of the corner $q\mathcal{Q}_{\mathbf{d}}q$ the induced representation of K has no nonzero fixed vectors.

Lemma 2.8 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \operatorname{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_n W_n(g) J_n^*)$ is the coordinate restriction of $q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n\mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed g, let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\left\|\widetilde{W}_n(g) - q_n U_n(g) q_n\right\| \longrightarrow 0.$$

For $g = 1$, the compression is already the identity q_n of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many q_n are nonzero. Retain and relabel those coordinates. Put $r_n = \operatorname{rank}(q_n)$, choose a unitary $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by J_n preserves operator norms and multiplicative defects. Thus (W_n) is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. □

Theorem 2.9 (normal Kazhdan radical theorem)**.** *Let G be countable and let $K \trianglelefteq G$ have property* (T) *with $K \le R_{\infty\to 2}(G)$. Then*

$$K \le \operatorname{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ. The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map from the original group, is such a representation of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K-fixed vectors. Normality of K gives

$$\Theta(g)p\Theta(g)^* = p \qquad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g)\,\mathrm{Fix}\,\Theta(K)$ and $\mathrm{Fix}\,\Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K, so every element of K has trivial corona image. Lemma 2.8 gives positive integers r_n and an operator norm asymptotic representation $W_n \colon G \to \mathcal{U}(r_n)$. Let $\widehat{\Theta} \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. Under the coordinatewise corner identifications, $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_{\mathbf{r}}$ faithfully. The Kazhdan projection of K under $\widehat{\Theta}$ is zero, so $\widehat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|}\sum_{s\in S}(\widehat{\Theta}(s) - 1)^*(\widehat{\Theta}(s) - 1)$$

the inequality

$$b \geq \frac{\kappa^2}{|S|}\,1.$$

The coordinate elements

$$b_n = \frac{1}{|S|}\sum_{s\in S}(W_n(s) - I_{r_n})^*(W_n(s) - I_{r_n})$$

represent b. Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields

$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from I_{r_n}. This contradicts $s_0 \in K \leq R_{\infty\to 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills K, as required. □

Proof of Theorem A. For every $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.7 gives

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G).$$

Normality of $R_{\infty\to 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G).$$

Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.9 now gives $K \leq \mathrm{Rad}_{\mathrm{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

A finite-dimensional linear representation over any field kills every generator of $\mathfrak{D}_G(L)$ by Theorem 2.3; its kernel is normal, so it contains $\mathfrak{D}_G(L)$. $\square$

3. The binary Leavitt group

This section proves Theorem B. The relation $s_0t_0 + s_1t_1 = 1$ of the binary Leavitt algebra gives an element $\tau \in H$ with $\tau L \tau^{-1} \leq L$ for a copy L of $\mathrm{EL}_3(R)$, and a commutator $d \neq 1$ in $\mathfrak{D}_H(L)$; since H is simple, $\mathfrak{D}_H(L) = H$, and Theorem A applies.

3.1. **The self-compression.** Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0t_0, \qquad q = s_1t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1qs_1 = 1. \tag{4}$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. Define a map $\Psi\colon M_3(R) \to M_3(R)$ by

$$\Psi(A) = qI_3 + s_0At_0. \tag{5}$$

Here s_0At_0 denotes the matrix $(s_0a_{ij}t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0q = 0$, and $t_0s_0 = 1$, all consequences of (2), show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0I_3t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so Ψ preserves the identity. Finally, $t_0\Psi(A)s_0 = A$, so Ψ is injective. Hence Ψ restricts to an injective group homomorphism $\mathrm{GL}_3(R) \to \mathrm{GL}_3(R)$.

To realize Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0I_3 & s_1t_0I_3 \\ 0 & t_1I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0I_3 & 0 \\ s_0t_1I_3 & s_1I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $XY = YX = I_6$, so $Y = X^{-1}$. Put

$$\tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{6}$$

A direct block multiplication gives

$$\tau\ \mathrm{diag}(A, I_9)\, \tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{7}$$

Lemma 3.1. *The matrix τ defined in (6) belongs to $\mathrm{EL}_{12}(R)$.*

Proof. Put $I = I_6$. The Whitehead factorization gives

(8)

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

If $B = (b_{rs})_{r,s=0}^5 \in M_6(R)$, then

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{r,6+s}(b_{rs}), \qquad \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{6+r,s}(b_{rs}).$$

In either product, distinct off-diagonal matrix units have product zero, so no cross-terms occur. Every factor on the right-hand side of (8) therefore belongs to $\mathrm{EL}_{12}(R)$. Since $Y = X^{-1}$, its left-hand side is τ, and the result follows. □

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \le H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 3.2. *The groups H and L have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \le L. \tag{9}$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \ne j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L. These matrices generate L, so (9) follows. □

3.2. Simplicity and the full MF radical.

Proposition 3.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. The ring R is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \mathrm{EL}_{12}(R)$, the subgroup $N \le \mathrm{GL}_{12}(R)$ is normalized by $\mathrm{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\mathrm{EL}_{12}(R, I) \le N \le C_{12}(R, I).$$

Simplicity of R gives $I = 0$ or $I = R$. If $I = R$, then $\mathrm{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \le C_{12}(R, 0) = Z(\mathrm{GL}_{12}(R))$. A matrix that commutes with every $e_{ij}(1)$ is a scalar matrix λI_{12}. Commutation with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $\lambda = 1$ and $N = 1$. Finally, $e_{01}(1) \ne 1$, so H is nontrivial. □

Proposition 3.4. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \ne 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

Proof. For $i, j \in \{0,1,2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L, so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{10}$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Its normal closure is a nontrivial normal subgroup of H, and hence is all of H by Proposition 3.3. □

Proof of Theorem B. The ring R is a finitely generated $\mathbb{F}_2$-algebra, so H is countable, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (9), the centrality of c relative to L, and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\!\langle d \rangle\!\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Proposition 3.3 shows that H is nontrivial and simple.

If M is MF and $f \colon H \to M$ is a homomorphism, compose f with a faithful corona embedding of M. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C^*_{\mathrm{r}}(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding $e \colon C^*_{\mathrm{r}}(H) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1-p)$ would embed H in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$. This contradicts the fact that H is not MF. □

4. Locality and finite certificates

The MF property is defined through sequences of models on all of G, but it is determined by finite data with a fixed separation constant. The reformulation is due to Korchagin [Kor]; this section derives the upper bound of Theorem C from it.

The following lemma is due to Korchagin [Kor, Proposition 7].

Lemma 4.1 (locality with fixed separation)**.** *A countable group G is MF if and only if for every finite set $F \subseteq G$ containing 1 and every $\varepsilon > 0$ there are $d \geq 1$ and a map $V \colon F \to \mathcal{U}(d)$ with $V(1) = 1$ such that*

$$\|V(gh) - V(g)V(h)\| \leq \varepsilon \quad \text{whenever } g, h, gh \in F,$$
$$\|V(g) - 1\| \geq 1 \quad \text{for } g \in F \setminus \{1\}.$$

Proof. If every finite $F \ni 1$ and every $\varepsilon > 0$ admit such a map, choose an exhaustion of G by finite sets F_n closed enough that $g, h, gh \in F_n$ for all large n, take a map for $(F_n, 1/n)$, and extend it by 1 outside F_n; the resulting sequence is an MF model. Conversely, given MF models V_n, Korchagin [Kor, Proposition 7] amplifies the separation: replacing $V_n(g)$ by the direct sum of its tensor powers $V_n(g)^{\otimes k}$, $k \le K$, multiplies the defects by at most K and raises $\limsup_n \|V_n(g) - 1\|$ above 1 for each of finitely many g, since an eigenvalue $e^{i\theta}$ of $V_n(g)$ with θ small has a power $e^{ik\theta}$ at angular distance at least $\pi/2$ from 1. A block direct sum of these amplified models over finitely many late indices, one for each $g \in F \setminus \{1\}$, then has defect at most ε on F and separation at least 1 from the identity on $F \setminus \{1\}$. ☐

Lemma 4.2 (marked closedness)**.** *The class of MF groups is closed in the space of marked groups. In particular, every finitely generated group that is not MF is a quotient of a finitely presented group that is not MF.*

Proof. An (F, ε)-model in the sense of Lemma 4.1 depends only on the multiplication table of F, that is, on which products gh with $g, h \in F$ lie in F and which elements of F are equal. If $G = \langle X \mid N \rangle$ is finitely generated and not MF, Lemma 4.1 gives a finite $F \ni 1$ and $\varepsilon > 0$ with no (F, ε)-model. Represent the elements of F by words in X; the finitely many equalities among the words and their pairwise products that hold in G are consequences of finitely many relators $N_0 \subseteq N$, and the inequalities hold in every marked group mapping onto G. Hence every marked group that agrees with G on the finitely many words involved carries a table isomorphic to that of F, and so is not MF; this is closedness. Applied to $\langle X \mid N_0 \rangle$, which maps onto G and has the same table, it gives the finitely presented quotient. ☐

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set $A \subseteq \mathbb{N}$ is Σ^0_1 if $A = \{x : \exists y\, R(x, y)\}$ for a decidable relation R, it is Π^0_1 if its complement is Σ^0_1, and Σ^0_2 and Π^0_2 are the classes $\{x : \exists y\, \forall z\, R(x, y, z)\}$ and $\{x : \forall y\, \exists z\, R(x, y, z)\}$ with R decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write G_P for the group presented by the code P. Let $\mathrm{MF}_{\mathrm{fp}}$ be the set of codes P such that G_P is MF, and $\mathrm{NONMF}_{\mathrm{fp}}$ its complement. Korchagin's Proposition 8 [Kor] is the finite-presentation form of Lemma 4.1; the certificates below add normal-closure witnesses for the trivial words, which is what makes the relation decidable.

Proposition 4.3 (finite certificates)**.** *There is a decidable relation $C(P, n, c)$ on triples of natural numbers such that a code P lies in* $\mathrm{MF}_{\mathrm{fp}}$ *if and only if for every n there is c with $C(P, n, c)$. Consequently* $\mathrm{MF}_{\mathrm{fp}} \in \Pi^0_2$ *and* $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma^0_2$.

Proof. Let $P = \langle x_1, \ldots, x_k \mid r_1, \ldots, r_m \rangle$. A *certificate at scale n* is a triple $c = (d, \ell, \pi)$ consisting of a dimension $d \ge 1$, a labelling ℓ of every reduced word of length at most n in $x_1^{\pm 1}, \ldots, x_k^{\pm 1}$ by one of the symbols T and S, and, for every word w labelled T, an expression of w as a product of conjugates of the relators and their inverses that freely reduces to w. Such an expression is checkable, and its existence forces $w = 1$ in G_P. Let $\Phi(P, n, c)$ be the statement that there are $U_1, \dots, U_k \in \mathcal{U}(d)$ with $\|r_i(U) - 1\| \le 2^{-n}$ for $i \le m$ and $\|w(U) - 1\| \ge 1/4$ for every word w labelled S, where $w(U)$ is the evaluation of w at $U_1, \dots, U_k$ with $x_j^{-1} \mapsto U_j^*$. Writing $U_j = A_j + iB_j$ with real matrices, unitarity is a system of polynomial equations, and a bound $\|X\| \le t$ or $\|X\| \ge t$ on a matrix whose entries are polynomials in the variables is a semialgebraic condition, since $\|X\| \le t$ holds precisely when $t^2 - X^*X$ is positive semidefinite. Thus $\Phi(P, n, c)$ is a sentence of the first-order theory of the ordered field of real numbers, and its truth is decidable by Tarski's theorem [Ta]. Let $C(P, n, c)$ hold when c is a well-formed certificate at scale n whose T-expressions check and $\Phi(P, n, c)$ holds. Then C is decidable.

Suppose that G_P is MF. Fix n, let $L = \max(n, |r_1|, \dots, |r_m|)$, let F be the set of images in G_P of all words of length at most L, together with their inverses, and let $\varepsilon = 2^{-n}/(4L)$. Lemma 4.1 gives $V\colon F \to \mathcal{U}(d)$ with $V(1) = 1$, defect at most ε on F, and $\|V(g) - 1\| \ge 1$ for $g \in F \setminus \{1\}$. Put $U_j = V(\bar{x}_j)$, where $\bar{w}$ denotes the image of a word w in G_P. For $g \in F$ we have $g^{-1} \in F$ and $\|V(g)V(g^{-1}) - 1\| \le \varepsilon$, so $\|V(g^{-1}) - V(g)^*\| \le \varepsilon$. For a word $w = y_1 \cdots y_\ell$ of length $\ell \le L$, all of whose prefixes have images in F, induction on ℓ gives $\|w(U) - V(\bar{w})\| \le 2\ell\varepsilon$. In particular $\|r_i(U) - 1\| \le 2L\varepsilon \le 2^{-n}$. Label a word w of length at most n by T if $\bar{w} = 1$, with a normal-closure expression as witness, and by S otherwise; for the latter, $\|w(U) - 1\| \ge \|V(\bar{w}) - 1\| - 2n\varepsilon \ge 1 - \frac{1}{2} \ge \frac{1}{4}$. So $C(P, n, c)$ holds for this certificate.

Conversely, suppose that for every n some certificate $c_n = (d_n, \ell_n, \pi_n)$ satisfies $C(P, n, c_n)$, and choose $U^{(n)} \in \mathcal{U}(d_n)^k$ witnessing $\Phi(P, n, c_n)$. For each $g \in G_P$ fix a reduced word w_g representing g, with w_1 the empty word, and put $V_n(g) = w_g(U^{(n)})$. Then $V_n(1) = 1$. For $g, h \in G_P$ the word $t = w_g w_h w_{gh}^{-1}$ is trivial in G_P, so it is freely equal to a product of A conjugates of relators and their inverses, for some A depending on g and h but not on n. Since evaluation is a homomorphism on the free group, since the operator norm is unitarily invariant, and since $\|xy - 1\| \le \|x - 1\| + \|y - 1\|$ for unitaries x, y,

$$\|V_n(g)V_n(h) - V_n(gh)\| = \left\| t(U^{(n)}) - 1 \right\| \le A\, 2^{-n} \longrightarrow 0.$$

For $g \ne 1$ the word w_g is nontrivial in G_P, so for $n \ge |w_g|$ the certificate c_n cannot label it T, hence labels it S, and $\|V_n(g) - 1\| \ge 1/4$. Thus $\limsup_n \|V_n(g) - 1\| \ge 1/4 > 0$, and G_P is MF.

The displayed equivalence exhibits $\mathrm{MF}_{\mathrm{fp}}$ as $\{P : \forall n\, \exists c\, C(P, n, c)\}$, a Π^0_2 set, and its complement as a Σ^0_2 set. □

5. HNN extensions with a corona conjugator

Sections 2 and 3 used a one-sided conjugation in a norm matrix corona to obstruct operator norm approximation. Theorem C also needs a positive statement: if the edge isomorphism of an HNN extension is implemented by a unitary of a norm matrix corona, then the HNN extension is MF. For $\theta = \mathrm{id}$ this is close to Shulman's theorem on central HNN extensions [Sh]; the version below assumes a tracial state vanishing off the identity, which is used to prove that the group embeds in the resulting algebra, and allows a nontrivial edge isomorphism. Two separate things are needed: a unitary of a corona implementing the edge isomorphism, which is finite-dimensional bookkeeping, and injectivity of the group in the resulting algebra, which comes from the trace and the left regular representation.

A countable group G *admits a tracial MF realization* (A, ρ, τ) if A is a separable unital MF C^*-algebra, $\rho\colon G \to \mathcal{U}(A)$ is a homomorphism, and τ is a tracial state on A with $\tau(\rho(g)) = 0$ for every $g \neq 1$. Since $\tau(1) = 1$, such a group embeds in $\mathcal{U}(A)$, so it is MF. Padding blocks with zeros embeds every norm matrix corona $\mathcal{Q}_{\mathbf{d}}$ isometrically into $\prod_n M_n(\mathbb{C})/\bigoplus_n M_n(\mathbb{C})$, so the MF algebras of the introduction are exactly the separable C^*-algebras that embed in the latter corona, as in [BK, Sh].

Lemma 5.1 (residually finite groups)**.** *Every countable residually finite group admits a tracial MF realization.*

Proof. Let G be countable and residually finite, and choose finite-index normal subgroups $N_1 \geq N_2 \geq \cdots$ with $\bigcap_k N_k = \{1\}$; this is possible because G is countable. Let $d_k = [G : N_k]$ and let $\lambda_k\colon G \to \mathcal{U}(d_k)$ be the left regular representation of G/N_k. In the corona $\mathcal{Q}_{\mathbf{d}}$ put $\rho(g) = [(\lambda_k(g))_k]$; then ρ is a homomorphism into the unitary group. Fix a free ultrafilter ω on $\mathbb{N}$ and set $\tau([(x_k)_k]) = \lim_\omega \mathrm{tr}_{d_k}(x_k)$, where tr_{d_k} is the normalized trace on $M_{d_k}(\mathbb{C})$; this is well defined because $|\mathrm{tr}_{d_k}(x_k)| \leq \|x_k\|$, and it is a tracial state on $\mathcal{Q}_{\mathbf{d}}$. If $g \neq 1$, then $g \notin N_k$ for all large k, so $\mathrm{tr}_{d_k}(\lambda_k(g)) = 0$ for all large k and $\tau(\rho(g)) = 0$. Let A be the C^*-algebra generated by $\rho(G)$ and the unit of $\mathcal{Q}_{\mathbf{d}}$; it is separable, unital, and MF, and $(A, \rho, \tau|_A)$ is a tracial MF realization of G. $\square$

Let G be a countable group, let $S \leq G$ be a subgroup, and let $\theta\colon S \to G$ be an injective homomorphism. The HNN extension

$$R = \langle G, t \mid tst^{-1} = \theta(s) \ (s \in S)\rangle \tag{11}$$

contains G by Britton's lemma [Bri].

Theorem 5.2 (HNN permanence with a corona conjugator)**.** *In the situation of* (11)*, suppose that G admits a tracial MF realization (A, ρ, τ), and that for some norm matrix corona $\mathcal{Q}$ there are an injective $*$-homomorphism $\iota\colon A \to \mathcal{Q}$ and a unitary $W \in \mathcal{Q}$ with*

$$W\, \iota\rho(s)\, W^* = \iota\rho(\theta(s)) \qquad (s \in S).$$

Then R admits a tracial MF realization. In particular, R is MF.

Proof. Write $D = C^*(\iota\rho(G)) \subseteq \mathcal{Q}$, a separable C^*-algebra with unit $p = \iota(1)$, and let $B_0 = C^*(\iota\rho(S))$ and $B_1 = C^*(\iota\rho(\theta(S)))$, both containing p.

Conjugation by W restricts to a $*$-isomorphism $\Theta\colon B_0 \to B_1$ carrying $\iota\rho(s)$ to $\iota\rho(\theta(s))$. Let U be the quotient of the full unital free product $D * C(\mathbb{T})$ by the closed ideal generated by the elements $ubu^* - \Theta(b)$, $b \in B_0$, where u is the canonical unitary generator of $C(\mathbb{T})$. We write d and u for the images of $d \in D$ and u in U. By construction, U has the following universal property: whenever $\pi\colon D \to E$ is a unital $*$-homomorphism into a unital C^*-algebra and $v \in \mathcal{U}(E)$ satisfies $v\pi(b)v^* = \pi(\Theta(b))$ for $b \in B_0$, there is a unique $*$-homomorphism $U \to E$ extending π and sending u to v. We first embed U in an amalgamated free product, which Shulman's criterion shows to be MF, and then construct the required tracial state on the copy of R inside U.

Step 1: U embeds in an amalgamated free product. This step follows a corner argument of Ueda [Ue]. Let $C = B_0 \oplus B_1$, $A_1 = M_2(D)$, and $A_2 = M_2(B_0)$, with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \qquad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let $P = A_1 *_C A_2$ be the full amalgamated free product, in which A_1 and A_2 embed [Sh, Section 2.1]. Write e_{ij} and f_{ij} for the matrix units of A_1 and A_2. Since $\iota_A(p, 0) = \iota_B(p, 0)$ and $\iota_A(0, p) = \iota_B(0, p)$, we have $e_{11} = f_{11} =: e$ and $e_{22} = f_{22}$ in P. Define $\pi_D\colon D \to ePe$ by $\pi_D(d) = \operatorname{diag}(d, 0)$, a unital $*$-homomorphism into the corner, and put $w = e_{12}f_{21} \in ePe$. Then $ww^* = e_{12}f_{22}e_{21} = e_{12}e_{22}e_{21} = e$ and $w^*w = f_{12}e_{22}f_{21} = f_{12}f_{22}f_{21} = e$, so w is a unitary of ePe. For $b \in B_0$ the element $\operatorname{diag}(b, 0)$ is $\iota_A(b, 0) = \iota_B(b, 0)$, so

$$\begin{aligned} w\pi_D(b)w^* &= e_{12}\big(f_{21}\operatorname{diag}(b, 0)f_{12}\big)e_{21} = e_{12}\,\iota_B(0, \Theta(b))\,e_{21} \\ &= e_{12}\,\iota_A(0, \Theta(b))\,e_{21} = \pi_D(\Theta(b)), \end{aligned}$$

where the middle equality uses $\iota_B(0, \Theta(b)) = \operatorname{diag}(0, b)$ in A_2. The universal property of U gives a $*$-homomorphism $\Phi\colon U \to ePe \subseteq P$ with $\Phi|_D = \pi_D$ and $\Phi(u) = w$. We prove that Φ is injective by constructing $\Psi\colon P \to M_2(U)$ with $\Psi\Phi(x) = \operatorname{diag}(x, 0)$. Let $\Psi_1\colon A_1 \to M_2(U)$ be the entrywise inclusion $D \subseteq U$, and let $\Psi_2\colon A_2 \to M_2(U)$ be the entrywise inclusion $B_0 \subseteq U$ followed by conjugation by $\operatorname{diag}(1, u)$. On C we have $\Psi_1(\iota_A(c_0, c_1)) = \operatorname{diag}(c_0, c_1)$ and $\Psi_2(\iota_B(c_0, c_1)) = \operatorname{diag}(c_0, u\Theta^{-1}(c_1)u^*) = \operatorname{diag}(c_0, c_1)$, so the pair induces a $*$-homomorphism $\Psi\colon P \to M_2(U)$. On generators, $\Psi\Phi(d) = \operatorname{diag}(d, 0)$ and $\Psi\Phi(u) = \Psi_1(e_{12})\Psi_2(f_{21}) = \operatorname{diag}(u, 0)$. Hence $\Psi\Phi$ is the injective $*$-homomorphism $x \mapsto \operatorname{diag}(x, 0)$ of U into $M_2(U)$, and Φ is injective.

Step 2: P is MF. By Shulman's criterion [Sh, Theorem 16], the full amalgamated free product $A_1 *_C A_2$ of separable C^*-algebras is MF as soon as there are injective $*$-homomorphisms φ_A and φ_B of A_1 and A_2 into one norm matrix corona with $\varphi_A \circ \iota_A = \varphi_B \circ \iota_B$. Identify $M_2(\mathcal{Q})$ with the norm matrix corona of the doubled dimension sequence, and let $\varphi_A\colon M_2(D) \to M_2(\mathcal{Q})$ be the entrywise inclusion and $\varphi_B\colon M_2(B_0) \to M_2(\mathcal{Q})$ the entrywise inclusion followed by conjugation by $\operatorname{diag}(1, W)$. Both are injective, and on C,

$$\varphi_B(\iota_B(c_0, c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0, c_1)),$$

because Θ is conjugation by W. Hence P is MF, and so are its C^*-subalgebras ePe and $\Phi(U) \cong U$.

Step 3: the tracial state. The functional $T = \tau \circ \iota^{-1}$ is a tracial state on $\iota(A) \supseteq D$ with $T(\iota\rho(g)) = 0$ for $g \neq 1$. In the GNS representation of D for T, the vectors $\xi_g = \iota\rho(g)\xi_T$, $g \in G$, are orthonormal, because $\langle \xi_g, \xi_h \rangle = T(\iota\rho(h^{-1}g))$. Their closed span is invariant under D, and on it $\iota\rho(g)$ acts as the left regular representation $\lambda_G(g)$ of G. Thus there is a $*$-homomorphism $\pi\colon D \to C^*_{\mathrm{r}}(G)$ with $\pi(\iota\rho(g)) = \lambda_G(g)$. Since $G \leq R$, the space $\ell^2(R)$ is a direct sum of copies of $\ell^2(G)$ indexed by the right cosets of G, so $\lambda_R|_G$ is a multiple of λ_G and the assignment $\lambda_G(g) \mapsto \lambda_R(g)$ extends to a $*$-homomorphism $C^*_{\mathrm{r}}(G) \to C^*_{\mathrm{r}}(R)$. Composing, we obtain $\sigma_0\colon D \to C^*_{\mathrm{r}}(R)$ with $\sigma_0(\iota\rho(g)) = \lambda_R(g)$. The pair $(\sigma_0, \lambda_R(t))$ satisfies the covariance relation $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$ for $b \in B_0$: on the generators $\iota\rho(s)$ both sides equal $\lambda_R(\theta(s))$, because $tst^{-1} = \theta(s)$ in R, and two $*$-homomorphisms that agree on generators agree on B_0. The universal property of U gives $\sigma\colon U \to C^*_{\mathrm{r}}(R)$ with $\sigma(\iota\rho(g)) = \lambda_R(g)$ and $\sigma(u) = \lambda_R(t)$.

Define $j\colon R \to \mathcal{U}(U)$ by $j(g) = \iota\rho(g)$ for $g \in G$ and $j(t) = u$. The defining relations of (11) hold in U, since $u\,\iota\rho(s)\,u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$, so j is a homomorphism, and $\sigma \circ j = \lambda_R$ is injective, so j is injective. Let $A' = C^*(j(R)) \subseteq U$, a separable unital MF C^*-algebra, and let $\tau' = \tau_R \circ \sigma|_{A'}$, where τ_R is the canonical trace $x \mapsto \langle x\delta_1, \delta_1\rangle$ of $C^*_{\mathrm{r}}(R)$. Then τ' is a tracial state on A' with $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$ for $r \neq 1$. Hence (A', j, τ') is a tracial MF realization of R. ☐

Injectivity of j follows from $\sigma \circ j = \lambda_R$, not from the covariant pair $(\iota\rho, W)$, which need not be faithful; in the applications below it is not.

Corollary 5.3 (central HNN extensions)**.** *If G admits a tracial MF realization and $S \leq G$ is any subgroup, then $\langle G, t \mid [t,s] = 1\ (s \in S)\rangle$ admits a tracial MF realization.*

Proof. Apply Theorem 5.2 with θ the inclusion of S, with ι any injective $*$-homomorphism of A into a norm matrix corona, and with $W = 1$. ☐

6. The complexity of recognizing MF groups

Write W_e for the domain of the e-th partial computable function and

$$\mathrm{INF} = \{e : W_e \text{ is infinite}\}, \qquad \mathrm{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is Π^0_2-complete and FIN is Σ^0_2-complete [So, Chapter IV]. We now combine the obstruction of Theorem B with the permanence of Theorem 5.2: this section constructs a computable map $e \mapsto \widehat{R}_e$ into finite presentations such that the presented group contains H when $e \in \mathrm{FIN}$ and satisfies the hypotheses of Theorem 5.2 when $e \in \mathrm{INF}$, so that it is MF if and only if $e \in \mathrm{INF}$. The construction of Bilanovic–Chubb–Roven gives a recursively presented group C_e that is trivial when $e \in \mathrm{INF}$ and contains H when $e \in \mathrm{FIN}$. We embed C_e in a three-generated recursively presented group $Q_e = F/N_e$, use a Mikhailova subgroup to detect N_e inside a finitely presented group, and apply two HNN extensions, in the manner of Higman's rope trick [Hig], to obtain $\widehat{R}_e$. This section proves that $\widehat{R}_e$ is not MF when $e \in \mathrm{FIN}$; Section 7 proves that it is MF when $e \in \mathrm{INF}$.

A recursive presentation of H.

Lemma 6.1 (a recursive presentation of H)**.** *The group $H = \mathrm{EL}_{12}(R)$ is finitely generated and has decidable word problem. In particular, it has a recursive presentation on finitely many generators.*

Proof. Let Y be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct i, j, k, and since every element of R is a sum of monomials in the generators s_0, s_1, t_0, t_1, induction on the length of a monomial shows that every $e_{ij}(a)$ lies in the subgroup generated by Y; there is always an index $k \notin \{i, j\}$ because $12 \geq 3$. Thus $H = \langle Y \rangle$. The monomials in s_0, s_1, t_0, t_1 that contain no subword $t_i s_j$ and no subword $s_1 t_1$ form the standard reduced $\mathbb{F}_2$-basis of R [Lev]. Products of reduced monomials are computed by the defining reductions, so equality in R is decidable. A word in $Y^{\pm 1}$ is therefore evaluated to an explicit 12×12 matrix over R, and it represents 1 in H exactly when that matrix is the identity. The set of words representing 1 is decidable, hence recursively enumerable, and $\langle Y \mid \text{that set} \rangle$ is a recursive presentation of H. ☐

The group C_e.

Lemma 6.2 (the group C_e)**.** *There is an algorithm that computes from e a recursive presentation, on a computable set of generators, of a group C_e such that C_e is trivial if $e \in \mathrm{INF}$ and H embeds in C_e if $e \in \mathrm{FIN}$.*

Proof. Apply the construction of Bilanovic–Chubb–Roven [BCR, Theorem 3.1] to the trivial group and H. Let $\langle Y \mid T \rangle$ be the recursive presentation of H from Lemma 6.1. Take generators $y_{i,\ell}$ for $i \geq 1$ and $\ell \in Y$, one copy Y_i of Y for each i, and as relators all words of T rewritten in each copy Y_i, together with every generator $y_{i,\ell}$ for which $i \leq |W_e|$. The last family is recursively enumerable uniformly in e: enumerate W_e, and whenever its i-th element appears, enumerate all of Y_i. The group presented is the free product of the copies $H_i \cong H$, $i \geq 1$, with the first $|W_e|$ copies killed. If W_e is infinite every copy is killed and C_e is trivial; if W_e is finite, C_e is the free product of the remaining copies and contains H. ☐

A three-generator embedding. Let $F = F(x, y, t)$ be the free group on x, y, t, and let $y_i = x^i y x^{-i}$ for $i \in \mathbb{Z}$. For a countable group C with a generating sequence $(c_i)_{i \geq 1}$, put $c_i = 1$ for $i \leq 0$ and

$$B(C) = \langle C * F(x, y),\ t \ \mid\ t y_i t^{-1} = c_i y_i \ (i \in \mathbb{Z}) \rangle. \tag{12}$$

Let $Q_+ = B(1) = \langle x, y, t \mid [t, y_i] = 1 \ (i \in \mathbb{Z}) \rangle$, let $q_+ \colon F \to Q_+$ be the quotient map, and let $N_+ = \ker q_+$.

Lemma 6.3 (the three-generator embedding)**.**

(1) $B(C)$ is an HNN extension of $C * F(x,y)$, so C embeds in $B(C)$; it is generated by x,y,t; and a recursive presentation of $B(C)$ on x,y,t is computable from a recursive presentation of C. We write $B(C) = F/N_C$ and $q_C \colon F \to B(C)$ for the quotient map.

(2) Killing C induces a surjection $B(C) \to Q_+$; hence $N_C \le N_+$, with equality when C is trivial.

(3) The assignment $x \mapsto (x_1,x_2)$, $y \mapsto (y,1)$, $t \mapsto (1,t)$ defines an injective homomorphism $j\colon Q_+ \to P = F(x_1,y) \times F(x_2,t)$. In particular, Q_+ is residually finite.

Proof. (1) The elements y_i, $i \in \mathbb{Z}$, freely generate the normal closure of y in $F(x,y)$ [LS, Chapter I]. The retraction $C * F(x,y) \to F(x,y)$ killing C sends $c_i y_i$ to y_i, so a nontrivial reduced word in the $c_i y_i$ maps to a nontrivial word in the y_i; hence the $c_i y_i$ freely generate a free subgroup, and $y_i \mapsto c_i y_i$ is an isomorphism between two free subgroups of the base. Thus (12) is an HNN extension, and the base, hence C, embeds by Britton's lemma. The relation $c_i = t y_i t^{-1} y_i^{-1}$ shows that x,y,t generate $B(C)$, and substituting this expression for c_i into the relators of C and into (12) produces a recursive presentation on x,y,t, uniformly in the presentation of C.

(2) The relators of (12) map to the relators $[t,y_i]$ of Q_+ when every c_i is sent to 1, and the map is the identity on x,y,t.

(3) The relators $[t,y_i]$ are sent to $[(1,t),(x_1^i y x_1^{-i},1)] = 1$, so j is a homomorphism. Let N_0 be the normal closure of $\{y,t\}$ in Q_+. Then $Q_+/N_0 = \langle x \rangle$ is infinite cyclic and the extension splits, so $Q_+ = N_0 \rtimes \langle x \rangle$. Rewriting the presentation of Q_+ over the transversal $\{x^n\}$ by the Reidemeister–Schreier method [LS, Chapter II] presents N_0 on the generators $y_n = x^n y x^{-n}$ and $t_m = x^m t x^{-m}$, $n,m \in \mathbb{Z}$, with relators the conjugates $x^m[t,y_n]x^{-m} = [t_m, y_{n+m}]$, that is, all commutators $[t_m,y_n]$. Hence $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$. Now $j(y_n) = (x_1^n y x_1^{-n}, 1)$ and $j(t_m) = (1, x_2^m t x_2^{-m})$, and the elements $x_1^n y x_1^{-n}$ freely generate the normal closure of y in $F(x_1,y)$, and likewise for t; so j restricted to N_0 is an isomorphism onto the product of these two normal closures. Finally $j(x) = (x_1,x_2)$ has infinite order and $j(x)^k \notin j(N_0)$ for $k \neq 0$, because the exponent sum of x_1 in $j(x^k n)$ is k for $n \in N_0$. Hence j is injective on $N_0 \rtimes \langle x \rangle$. Free groups are residually finite and residual finiteness passes to direct products and subgroups, so Q_+ is residually finite. □

For $e \in \mathbb{N}$ let $Q_e = B(C_e) = F/N_e$ with C_e the group of Lemma 6.2, and write $q_e \colon F \to Q_e$. By Lemmas 6.2 and 6.3, a recursive presentation of Q_e on x,y,t is computable from e; $N_e \le N_+$; if $e \in \mathrm{INF}$ then $Q_e = Q_+$, $N_e = N_+$ and $q_e = q_+$; and if $e \in \mathrm{FIN}$ then $H \le C_e \le Q_e$.

Detecting N_e inside a finitely presented group.

Lemma 6.4 (Higman embedding and the Mikhailova subgroup)**.** *There is an algorithm that computes from e a finite presentation $\langle X_e \mid R_e \rangle$ of a group Λ_e, words $w_x, w_y, w_t \in F(X_e)$, and a finite generating set of a subgroup*

$M_e \le F(X_e) \times F(X_e)$, such that $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$ induces an embedding $Q_e \to \Lambda_e$, and $M_e = \{(u,v) : u = v \text{ in } \Lambda_e\}$. Consequently, with

$$K_e^0 = F \times F(X_e) \times F(X_e), \qquad i_e(f) = (f, w_f, 1), \qquad L_e^0 = F \times M_e,$$

where w_f is the image of f under $x \mapsto w_x$, $y \mapsto w_y$, $t \mapsto w_t$, the map $i_e \colon F \to K_e^0$ is injective and $i_e(f) \in L_e^0$ if and only if $f \in N_e$.

Proof. Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of Q_e. Following Mikhailova [Mih], let M_e be the subgroup of $F(X_e) \times F(X_e)$ generated by the pairs (z,z), $z \in X_e$, and $(r,1)$, $r \in R_e$. If $(u,v) \in M_e$ then $u = v$ in Λ_e, since this holds for the generators. Conversely, if $u = v$ in Λ_e, then uv^{-1} lies in the normal closure of R_e, so it is a product of conjugates $gr^{\pm 1}g^{-1}$ with $r \in R_e$, and $(gr^{\pm 1}g^{-1}, 1) = (g,g)(r^{\pm 1},1)(g,g)^{-1} \in M_e$; hence $(u,v) = (uv^{-1},1)(v,v) \in M_e$. The first coordinate makes i_e injective, and $(f, w_f, 1) \in F \times M_e$ if and only if $w_f = 1$ in Λ_e, if and only if $q_e(f) = 1$ because the embedding is injective, if and only if $f \in N_e$. □

The two HNN extensions. Let $P = F(x_1, y) \times F(x_2, t)$ and $j \colon Q_+ \to P$ be as in Lemma 6.3, and put

$$K^{\mathrm{g}} = F \times P, \qquad L^{\mathrm{g}} = \{(f,\, jq_+(f)) : f \in F\},$$

and define

$$K_e = K_e^0 \times K^{\mathrm{g}}, \qquad L_e = L_e^0 \times L^{\mathrm{g}}, \qquad i \colon F \to K_e, \quad i(f) = \big(i_e(f), (f,1)\big).$$

Let

$$\Gamma_e = \langle K_e,\ v \mid [v,\ell] = 1 \ (\ell \in L_e) \rangle, \qquad S_e = \langle i(F),\ v\, i(F)\, v^{-1} \rangle \le \Gamma_e. \tag{13}$$

Lemma 6.5 (the first HNN extension)**.**

(1) K_e is a direct product of free groups of finite rank, hence finitely presented and residually finite; L_e is finitely generated; i is injective; and $i(F) \cap L_e = i(N_e)$.

(2) Γ_e is finitely presented, and a finite presentation is computable from e. The natural map from the amalgamated free product $F *_{N_e} F$, in which $n \in N_e$ in the first copy is identified with n in the second copy, onto S_e, sending the first copy to $i(F)$ and the second to $v\, i(F)\, v^{-1}$, is an isomorphism.

(3) There is a homomorphism $\tau_e \colon S_e \to Q_e$ with $\tau_e(i(f)) = q_e(f)$ and $\tau_e(v\, i(f)\, v^{-1}) = 1$ for $f \in F$.

Proof. (1) $K_e = F \times F(X_e) \times F(X_e) \times F \times P$ is a product of six free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup L_e is generated by the basis of the first factor, the generators of M_e from Lemma 6.4, and the elements $(a, jq_+(a))$ for

$a \in \{x, y, t\}$, so it is finitely generated. Injectivity of i is clear from the first coordinate. An element $i(f)$ lies in L_e if and only if $i_e(f) \in L_e^0$ and $(f, 1) \in L^{\mathrm{g}}$; the first holds if and only if $f \in N_e$ by Lemma 6.4, and the second if and only if $jq_+(f) = 1$, that is, $f \in N_+$. Since $N_e \le N_+$, the intersection is $i(N_e)$.

(2) The presentation (13) is finite once $[v, \ell] = 1$ is imposed only for the finitely many generators ℓ of L_e, and it is computable from e because K_e, L_e and i are. It is an HNN extension of K_e with both edge groups equal to L_e and the identity as edge isomorphism. The map from $F *_{N_e} F$ to S_e is well defined because $i(n) \in L_e$ commutes with v for $n \in N_e$. A reduced word of the amalgamated free product alternates between elements of the two copies of F lying outside N_e; its image is a word $i(f_0)\, v\, i(f_1)\, v^{-1}\, i(f_2) \cdots$ in which every letter between consecutive occurrences of $v^{\pm 1}$ is some $i(f_k)$ with $f_k \notin N_e$, hence $i(f_k) \notin L_e$ by (1). Such a word contains no pinch $v^{\pm 1} \ell v^{\mp 1}$ with $\ell \in L_e$, so it is nontrivial in Γ_e by Britton's lemma.

(3) Define τ_e on $F *_{N_e} F$ by q_e on the first copy and by the trivial map on the second; both kill N_e, so the definition is consistent on the amalgamated subgroup, and (2) identifies $F *_{N_e} F$ with S_e. $\square$

The second HNN extension is

$$R_e = \langle \Gamma_e \times Q_e,\ u \ \mid\ u(s, 1)u^{-1} = (s, \tau_e(s))\ (s \in S_e) \rangle, \tag{14}$$

whose edge maps $s \mapsto (s, 1)$ and $s \mapsto (s, \tau_e(s))$ are injective homomorphisms of S_e into $\Gamma_e \times Q_e$; so $\Gamma_e \times Q_e$ embeds in R_e by Britton's lemma. The presentation (14) lists the recursively enumerable relators of Q_e. Its finite replacement is

$$\widehat{R}_e = \Big\langle \Gamma_e \times F,\ u\ \Big|\ \begin{aligned}[t] & u(i(a), 1)u^{-1} = (i(a), a), \\ & u(v\, i(a)\, v^{-1}, 1)u^{-1} = (v\, i(a)\, v^{-1}, 1) \quad (a \in \{x, y, t\}) \end{aligned} \Big\rangle, \tag{15}$$

where $\Gamma_e \times F$ is equipped with the finite presentation of Lemma 6.5(2), the basis x, y, t of F, and the commutation relators between the two factors.

Lemma 6.6 (the finite presentation)**.** *The groups $\widehat{R}_e$ and R_e are isomorphic by the map that is the identity on Γ_e and on u and sends F onto Q_e by q_e. A code of the finite presentation (15) is computable from e.*

Proof. Both maps $f \mapsto u(i(f), 1)u^{-1}$ and $f \mapsto (i(f), f)$ are homomorphisms $F \to \widehat{R}_e$, and they agree on the basis, so $u(i(f), 1)u^{-1} = (i(f), f)$ for all $f \in F$; likewise $u(v\, i(f)\, v^{-1}, 1)u^{-1} = (v\, i(f)\, v^{-1}, 1)$ for all f. If $n \in N_e$, then $i(n) \in L_e$ by Lemma 6.5(1), so $v\, i(n)\, v^{-1} = i(n)$ in Γ_e, and comparing the two relations gives $(i(n), n) = (i(n), 1)$, that is, $(1, n) = 1$ in $\widehat{R}_e$. Hence the factor F of $\widehat{R}_e$ factors through $Q_e = F/N_e$, and the relations of (15) become $u(s, 1)u^{-1} = (s, \tau_e(s))$ for s in the generating set $\{i(a),\ v\, i(a)\, v^{-1} : a \in \{x, y, t\}\}$ of S_e; both sides define homomorphisms on S_e, so the relations hold for all $s \in S_e$. This defines a homomorphism

$R_e \to \widehat{R}_e$; conversely, the relations of (15) hold in R_e after $F \to Q_e$, because $\tau_e(i(a)) = q_e(a)$ and $\tau_e(v\,i(a)\,v^{-1}) = 1$. The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 6.4 and 6.5. □

Lemma 6.7 (the case $e \in \mathrm{FIN}$)**.** *If $e \in \mathrm{FIN}$, then H embeds in $\widehat{R}_e$, and $\widehat{R}_e$ is not MF.*

Proof. For $e \in \mathrm{FIN}$ we have $H \le C_e \le Q_e \le \Gamma_e \times Q_e \le R_e \cong \widehat{R}_e$ by Lemmas 6.2, 6.3(1), 6.6 and Britton's lemma for (14). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and H is not MF by Theorem B. □

7. The case $e \in \mathrm{INF}$

We now prove that $\widehat{R}_e$ is MF for $e \in \mathrm{INF}$ and deduce Theorem C. The tracial MF realization of R_e is constructed inside a norm reduced product of MF algebras, so we first record that separable subalgebras of such products are MF. For C^*-algebras B_n, $n \ge 1$, write $\prod_n B_n / \bigoplus_n B_n$ for the quotient of the bounded sequences by the sequences with $\|x_n\| \to 0$; its norm is $\limsup_n \|x_n\|$.

Lemma 7.1 (reduced products of MF algebras)**.** *If every B_n is MF, then every separable C^*-subalgebra of $\prod_n B_n / \bigoplus_n B_n$ is MF. Moreover, $M_k(B)$ is MF whenever B is.*

Proof. The second assertion holds because M_k of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by k. For the first, let A be a separable C^*-subalgebra, and choose a sequence $(a_i)_{i\ge1}$ dense in A, with lifts $(a_i^{(n)})_n$. Choose for each n an injective $*$-homomorphism η_n of B_n into a norm matrix corona, and lifts $(a_i^{(n,k)})_k$ of $\eta_n(a_i^{(n)})$ with $\sup_k \left\|a_i^{(n,k)}\right\| \le \left\|a_i^{(n)}\right\| + 1$. Enumerate the noncommutative $*$-polynomials with coefficients in $\mathbb{Q}(i)$ as $p_1, p_2, \ldots$, and let $\mathcal{P}_n$ be the set of those among $p_1, \ldots, p_n$ that involve only $a_1, \ldots, a_n$, choosing $p_1 = 0$ so that $\mathcal{P}_n$ is nonempty. For $p \in \mathcal{P}_n$ the number $\left\|p(a^{(n)})\right\|$ equals $\limsup_k \left\|p(a^{(n,k)})\right\|$, since η_n is isometric. Hence there is K_n such that $\left\|p(a^{(n,k)})\right\| \le \left\|p(a^{(n)})\right\| + 1/n$ for all $k \ge K_n$ and $p \in \mathcal{P}_n$, and for each $p \in \mathcal{P}_n$ there is $k_p \ge K_n$ with $\left\|p(a^{(n,k_p)})\right\| \ge \left\|p(a^{(n)})\right\| - 1/n$. Put

$$b_i^{(n)} = \bigoplus_{p \in \mathcal{P}_n} a_i^{(n,k_p)},$$

a block diagonal matrix. For $p \in \mathcal{P}_n$ the norm of $p(b^{(n)})$ is the maximum of $\left\|p(a^{(n,k_{p'})})\right\|$ over $p' \in \mathcal{P}_n$, which lies between $\left\|p(a^{(n)})\right\| - 1/n$ and $\left\|p(a^{(n)})\right\| + 1/n$. Every polynomial belongs to $\mathcal{P}_n$ for all large n, so $\limsup_n \left\|p(b^{(n)})\right\| = \limsup_n \left\|p(a^{(n)})\right\| = \|p(a)\|$ for every p. Therefore $a_i \mapsto [(b_i^{(n)})_n]$ is well defined and isometric on the $*$-algebra over $\mathbb{Q}(i)$ generated by the a_i, which is dense in A, and it extends to an isometric $*$-homomorphism of A into a norm matrix corona. $\square$

Lemma 7.2 (tensor synchronization)**.** *Let Γ be a countable group with a tracial MF realization (A_1, ρ_1, τ_1), let Q be a countable group with homomorphisms $\beta_n\colon Q \to B_n$ to finite groups such that every $q \neq 1$ satisfies $\beta_n(q) \neq 1$ for all large n, let $S \leq \Gamma$, and let $\tau\colon S \to Q$ be a homomorphism. Suppose that for every n there is a homomorphism $\lambda_n\colon \Gamma \to G_n$ to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

Then there are a separable unital MF C^*-algebra A, a homomorphism $V\colon \Gamma \times Q \to \mathcal{U}(A)$, a tracial state T on A with $T(V(g,q)) = 0$ for $(g,q) \neq (1,1)$, and a unitary $W \in A$ with $WV(s,1)W^* = V(s,\tau(s))$ for all $s \in S$. Consequently

$$\langle \Gamma \times Q,\ u \ \mid\ u(s,1)u^{-1} = (s,\tau(s))\ (s \in S)\rangle$$

admits a tracial MF realization, and in particular is MF.

Proof. Let E_n be the image of $(\lambda_n, \beta_n)\colon \Gamma \times Q \to G_n \times B_n$, a finite group of order k_n, and let L_n be its left regular representation on $\ell^2(E_n)$. Put $B_n' = A_1 \otimes M_{k_n}(\mathbb{C})$, which is MF by Lemma 7.1, let $E = \prod_n B_n' / \bigoplus_n B_n'$, and define

$$V(g,q) = \big[\big(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q))\big)_n\big], \qquad T([(x_n)_n]) = \lim_{\omega} (\tau_1 \otimes \mathrm{tr}_{k_n})(x_n),$$

for a free ultrafilter ω. Each coordinate of V is a homomorphism into a unitary group, so V is a homomorphism, and T is a tracial state on E because $|(\tau_1 \otimes \mathrm{tr}_{k_n})(x_n)| \leq \|x_n\|$. If $g \neq 1$ then $\tau_1(\rho_1(g)) = 0$, so $T(V(g,q)) = 0$; if $g = 1$ and $q \neq 1$ then $\beta_n(q) \neq 1$ for all large n, the regular representation of a finite group has trace zero off the identity, and again $T(V(1,q)) = 0$.

The two homomorphisms $s \mapsto (\lambda_n(s), 1)$ and $s \mapsto (\lambda_n(s), \beta_n\tau(s))$ from S to E_n have the same kernel, namely $\ker(\lambda_n|_S)$, by the hypothesis. Their images are therefore isomorphic subgroups of E_n of the same order, isomorphic via $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n\tau(s))$. The restriction of L_n to a subgroup of E_n is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary $W_n \in M_{k_n}(\mathbb{C})$ with $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n\tau(s))$ for all $s \in S$. Put $W = [(1 \otimes W_n)_n] \in E$; then $WV(s,1)W^* = V(s,\tau(s))$ for $s \in S$. Let A be the C^*-algebra generated by $V(\Gamma \times Q)$ and W; it is separable and unital, and MF by Lemma 7.1. The triple $(A, V, T|_A)$ is a tracial MF realization of $\Gamma \times Q$. For the last assertion, apply Theorem 5.2 to this realization. Let ι be an injective $*$-homomorphism of A into a norm matrix corona and put $p = \iota(1)$. The element $\widetilde{W} = \iota(W) + (1-p)$ is a unitary of the full corona and implements the required covariance on $\iota(A)$, so it is the required conjugator. $\square$

Lemma 7.3 (the case $e \in \mathrm{INF}$)**.** *If $e \in \mathrm{INF}$, then $\widehat{R}_e$ admits a tracial MF realization; in particular, it is MF.*

Proof. Let $e \in \mathrm{INF}$, so that $Q_e = Q_+$, $N_e = N_+$, $q_e = q_+$, and $\tau_e(i(f)) = q_+(f)$, $\tau_e(v\,i(f)\,v^{-1}) = 1$. By Lemma 6.6 it suffices to show that R_e admits a tracial MF realization, and by Lemma 7.2 applied to $\Gamma = \Gamma_e$, $Q = Q_+$, $S = S_e$ and $\tau = \tau_e$, it suffices to produce the data of that lemma.

The group K_e is residually finite by Lemma 6.5(1), hence admits a tracial MF realization by Lemma 5.1, and Γ_e is the central HNN extension of K_e over L_e, hence admits one by Corollary 5.3.

The group $P = F(x_1, y) \times F(x_2, t)$ is residually finite; choose finite-index normal subgroups $P_1 \geq P_2 \geq \cdots$ of P with trivial intersection, let $r_n \colon P \to C_n = P/P_n$, and put $\beta_n = r_n \circ j \colon Q_+ \to C_n$ with j the embedding of Lemma 6.3(3). Since j is injective and the P_n decrease to the trivial group, every $h \neq 1$ has $\beta_n(h) \neq 1$ for all large n.

Let $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$, where σ of order two exchanges the coordinates. Define λ_n on $K_e = K_e^0 \times (F \times P)$ by killing K_e^0 and sending (f, p) to $(r_n j q_+(f),\, r_n(p))$, and put $\lambda_n(v) = \sigma$. This respects the relations of (13): the factor L_e^0 of L_e lies in K_e^0 and is killed, and an element $(f, jq_+(f))$ of L^{g} is sent to $(r_n j q_+(f), r_n j q_+(f))$, which lies on the diagonal and commutes with σ. Hence $\lambda_n \colon \Gamma_e \to G_n$ is a homomorphism, and on the generators of S_e,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \qquad \lambda_n(v\,i(f)\,v^{-1}) = (1, r_n j q_+(f)).$$

Let $\pi_0, \pi_1 \colon S_e \to Q_+$ be the homomorphisms that are q_+ on one copy of F in $S_e \cong F *_{N_+} F$ and trivial on the other, as in Lemma 6.5(3); thus $\pi_0 = \tau_e$. The displayed formulas show that $\lambda_n|_{S_e} = (r_n j \pi_0,\, r_n j \pi_1)$, both sides being homomorphisms that agree on generators. Therefore

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \tau_e),$$

which is the hypothesis of Lemma 7.2. □

Proof of Theorem C. By Lemma 6.6 the map $e \mapsto \widehat{R}_e$ is computable, by Lemma 7.3 the presented group is MF for $e \in \mathrm{INF}$, and by Lemma 6.7 it is not MF for $e \in \mathrm{FIN}$. Hence INF reduces to $\mathrm{MF}_{\mathrm{fp}}$ and FIN reduces to $\mathrm{NONMF}_{\mathrm{fp}}$ under computable many-one reductions. Since INF is Π^0_2-complete and FIN is Σ^0_2-complete, and since $\mathrm{MF}_{\mathrm{fp}} \in \Pi^0_2$ and $\mathrm{NONMF}_{\mathrm{fp}} \in \Sigma^0_2$ by Proposition 4.3, the two sets are complete at their levels. A Π^0_2-complete set is not Σ^0_2 and a Σ^0_2-complete set is not Π^0_2, so neither set is recursively enumerable, and in particular neither is decidable. □

The homomorphism V of Lemma 7.2 does not extend faithfully to R_e in the corona: the finite maps λ_n are trivial on the Mikhailova factor, so W commutes with $V(\gamma, 1)$ for every $\gamma \in K_e^0$, although u does not commute with $(\gamma, 1)$ in R_e. Injectivity is obtained in Theorem 5.2 from the universal algebra. The following questions are open. It is not known whether the models of $\widehat{R}_e$ can be chosen to converge to $C^*_{\mathrm{r}}(\widehat{R}_e)$ in operator norm, so that the reduced C^*-algebra itself is MF; the argument produces a tracial state on an abstract MF algebra, not strong convergence. It is not known whether $\widehat{R}_e$ is hyperlinear for $e \in \mathrm{INF}$, which would place the recognition problem for hyperlinearity at the same level. Finally, it is not known whether an analogous family of finitely presented groups exists for soficity; the construction of Lemma 7.2 has no counterpart for permutation models.

Acknowledgments

The author thanks Francesco Fournier-Facio for early advice and detailed comments on earlier drafts, and Alon Dogon for organizing the ensuing discussion and for comments on the surrounding literature. The author also thanks Tatiana Shulman for clarifying the terminology and historical context of the MF question, and Caleb Eckhardt for discussions of MF representations and matrix norms.

Use of AI and formal methods

Claude Fable 5 and GPT-5.6 Sol assisted with mathematical exploration, proof development, literature searches, Lean formalization, drafting, and editing. The author reviewed their output and is responsible for the manuscript. Lean 4 source is available at `https://github.com/SauersML/group-approximation`. The development verifies Theorems A and B, the finite certificates of Proposition 4.3, and the deduction of Theorem C from a computable family of finite presentations that is MF exactly on INF; the construction of that family in Section 6 and Theorem 5.2 are proved in the article. The repository also contains machine-checked results not used here, on MF radicals, the recognition of approximation properties, and related group constructions, with an indexed list of statements and declarations.

References

[AA] G. Abrams and G. Aranda Pino, *Purely infinite simple Leavitt path algebras*, J. Pure Appl. Algebra **207** (2006), no. 3, 553–563, doi:10.1016/j.jpaa.2005.10.010.

[Ad] S. I. Adian, *Algorithmic unsolvability of problems of recognition of certain properties of groups*, Dokl. Akad. Nauk SSSR **103** (1955), 533–535.

[AW] C. A. Akemann and M. E. Walter, *Unbounded negative definite functions*, Canad. J. Math. **33** (1981), no. 4, 862–871.

[AT] V. Alekseev and A. Thom, *Centralizers of sofic approximations of Kazhdan groups*, preprint, 2026, arXiv:2608.05362.

[AC] G. Aranda Pino and K. Crow, *The center of a Leavitt path algebra*, Rev. Mat. Iberoam. **27** (2011), no. 2, 621–644, doi:10.4171/RMI/649.

[Ara] P. Ara, *The exchange property for purely infinite simple rings*, Proc. Amer. Math. Soc. **132** (2004), no. 9, 2543–2547, doi:10.1090/S0002-9939-04-07369-1.

[BCR] I. Bilanovic, J. Chubb, and S. Roven, *Detecting properties from descriptions of groups*, Arch. Math. Logic **59** (2020), no. 3–4, 293–312, doi:10.1007/s00153-019-00690-x.

[BHV] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan's Property (T)*, Cambridge University Press, 2008.

[BDL] B. Bachner, A. Dogon, and A. Lubotzky, *On L^1-approximation of groups*, J. Algebra **702** (2026), 235–243, doi:10.1016/j.jalgebra.2026.04.041, arXiv:2508.17392.

[BK94] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional C^*-algebras*, in *C^*-Algebren*, Tagungsbericht 6/1994, Mathematisches Forschungsinstitut Oberwolfach, 1994, p. 2.

[BK] B. Blackadar and E. Kirchberg, *Generalized inductive limits of finite-dimensional C^*-algebras*, Math. Ann. **307** (1997), no. 3, 343–380, doi:10.1007/s002080050039.

[Bri] J. L. Britton, *The word problem*, Ann. of Math. (2) **77** (1963), 16–32.

[CDE] J. R. Carrión, M. Dadarlat, and C. Eckhardt, *On groups with quasidiagonal C^*-algebras*, J. Funct. Anal. **265** (2013), no. 1, 135–152, doi:10.1016/j.jfa.2013.04.004.

[Dad] M. Dadarlat, *Obstructions to matricial stability of discrete groups and almost flat K-theory*, Adv. Math. **384** (2021), Paper No. 107722, doi:10.1016/j.aim.2021.107722.

[DGLT] M. De Chiffre, L. Glebsky, A. Lubotzky, and A. Thom, *Stability, cohomology vanishing, and nonapproximable groups*, Forum Math. Sigma **8** (2020), Paper No. e18, doi:10.1017/fms.2020.5.

[EJZ] M. Ershov and A. Jaikin-Zapirain, *Property* (T) *for noncommutative universal lattices*, Invent. Math. **179** (2010), 303–347, doi:10.1007/s00222-009-0218-2.

[GH] I. Goldbring and B. Hart, *The universal theory of the hyperfinite* II_1 *factor is not computable*, Bull. Symbolic Logic **30** (2024), no. 2, 181–198, doi:10.1017/bsl.2024.7, arXiv:2006.05629.

[Hig] G. Higman, *Subgroups of finitely presented groups*, Proc. Roy. Soc. London Ser. A **262** (1961), 455–475.

[JNVWY] Z. Ji, A. Natarajan, T. Vidick, J. Wright, and H. Yuen, $\mathrm{MIP}^* = \mathrm{RE}$, Commun. ACM **64** (2021), no. 11, 131–138, doi:10.1145/3485628, arXiv:2001.04383.

[Kor] A. I. Korchagin, *The MF-property for countable discrete groups*, Math. Notes **102** (2017), no. 2, 198–211, doi:10.1134/S0001434617070227, arXiv:1704.06906.

[Lev] W. G. Leavitt, *The module type of a ring*, Trans. Amer. Math. Soc. **103** (1962), 113–130, doi:10.1090/S0002-9947-1962-0132764-6.

[Lem] S. Lempp, *The computational complexity of torsion-freeness of finitely presented groups*, Bull. Austral. Math. Soc. **56** (1997), 273–277.

[LS] R. C. Lyndon and P. E. Schupp, *Combinatorial group theory*, Classics in Mathematics, Springer-Verlag, Berlin, 2001.

[LO] A. Lubotzky and I. Oppenheim, *Non p-norm approximated groups*, J. Anal. Math. **141** (2020), no. 1, 305–321, doi:10.1007/s11854-020-0119-2.

[KT] G. Kun and A. Thom, *Inapproximability of actions and Kazhdan's property* (T), preprint, 2019, arXiv:1901.03963.

[Mik] V. H. Mikaelian, *An explicit algorithm for the Higman embedding theorem*, preprint, 2025, arXiv:2507.04347.

[Mih] K. A. Mihailova, *The occurrence problem for direct products of groups*, Dokl. Akad. Nauk SSSR **119** (1958), 1103–1105.

[OAI] OpenAI, *Nonsofic groups exist*, in *Ten advances in mathematics and theoretical computer science*, Chapter 3, 78–95, 2026, https://cdn.openai.com/pdf/ten-proofs-oai.pdf.

[Pre] R. Preusser, *On general linear groups over exchange rings*, Linear and Multilinear Algebra **70** (2022), no. 4, 705–713, doi:10.1080/03081087.2020.1743636.

[Ra] M. O. Rabin, *Recursive unsolvability of group theoretic problems*, Ann. of Math. (2) **67** (1958), 172–194.

[Sh] T. Shulman, *The MF property for amalgamated free products*, preprint, 2026, arXiv:2603.13564.

[So] R. I. Soare, *Recursively enumerable sets and degrees*, Perspectives in Mathematical Logic, Springer-Verlag, Berlin, 1987.

[Ta] A. Tarski, *A decision method for elementary algebra and geometry*, 2nd ed., University of California Press, Berkeley, 1951.

[Th] A. Thom, *Finitary approximations of groups and their applications*, in *Proceedings of the International Congress of Mathematicians—Rio de Janeiro 2018*, Vol. III: Invited Lectures, World Scientific, Hackensack, NJ, 2018, 1779–1799, doi:10.1142/9789813272880_0117, arXiv:1712.01052.

[Ue] Y. Ueda, *Remarks on HNN extensions in operator algebras*, Illinois J. Math. **52** (2008), no. 3, 705–725.